

Doomscrolling and Mental Health

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Background

Main Topic

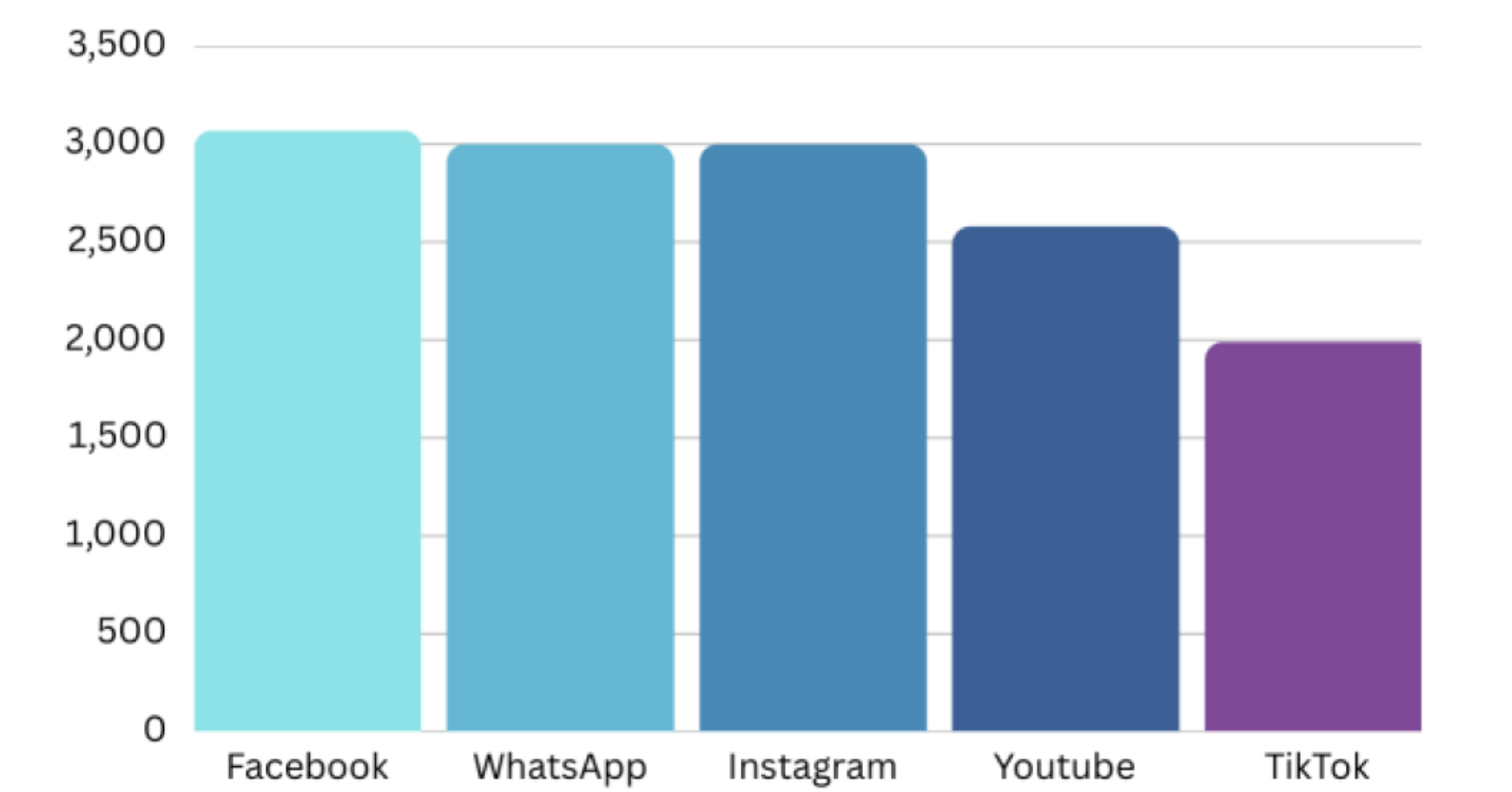
Better understand the extent of social media dependency among students and to identify patterns that may contribute to unhealthy usage behavior.

Why do we care?

Social media has become an essential part of student life, offering opportunities for communication, entertainment, and learning. Excessive use can lead to addiction, reduced academic performance, sleep disturbances, and mental health challenges.

As of 2025, roughly 60% of the global population (4.8 billion) are registered on social media, with the average amount of social media screen time being 2 and half hours.

If we add all of that together that equates to 11.5 billion hours daily!
(University of Maine 2023)



"Social Media Statistics Details - Undiscovered Maine - University of Maine." Undiscovered Maine, <https://umaine.edu/undiscoveredmaine/small-business/resources/marketing-for-small-business/social-media-tools/social-media-statistics-details/>. Accessed 6 Mar. 2026.

The Data

Overview

The dataset had 13 total variables, but we decided to only use data from 4.

- Mental Health Score
- Average Daily Usage
- Average Sleep Hours
- Conflicts over Social Media

The Plan

Use linear and logistic regression with Mental Health Score as the response variable and our predictors being Daily Usage, Sleep Hours, and Conflicts.

Linear Regression: creates a line that models our data and makes predictions for our response variable

Logistic Regression: creates a S-curve that would model our data to make predictions.

Regression → Classification

Logistic regression requires a binary response variable and Mental Health is not that. Thus, we chose to define 0 and 1 as low(score <= 6) and high mental health(score > 6). This transformed our response variable into a binary variable.

Simple Linear Regression

Overview

Our predictors: average hours spent on social media, hours of sleep per night, and the number of conflicts on social media.

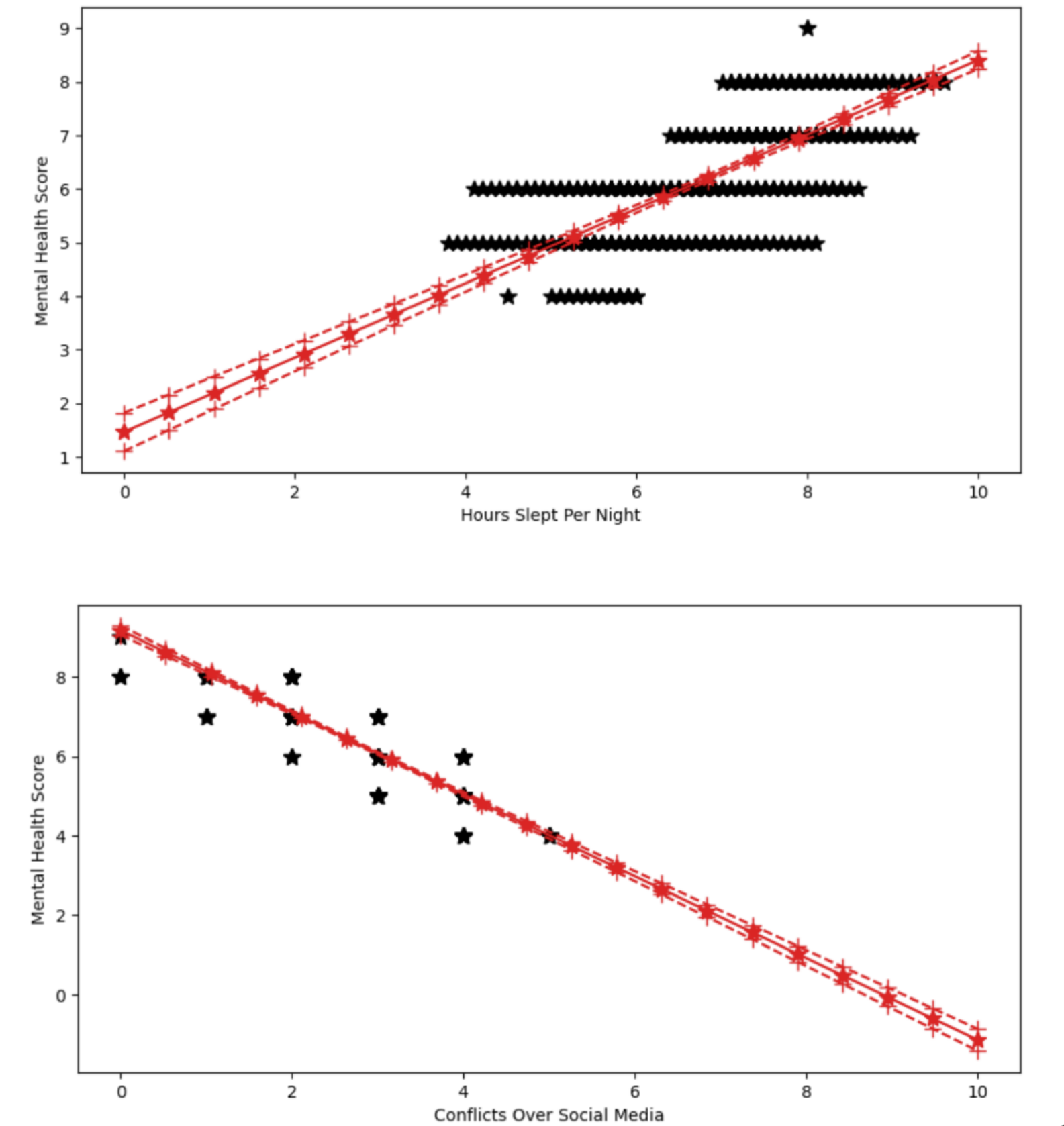
Output: Mental Health Score

Results

All three simple regression models contained a p-value less than 0.001 for each predictor.

Average daily hours spent on social media showed a clear negative relationship with mental health. Each hour spent online was a 0.7 point decrease in the mental health score. Hours of sleep per night showed a positive relationship with a 0.7 point increase in the mental health score. Conflicts on social media had the strongest relationship with mental health, with approximately a 1-point decrease in the mental health score.

The lowest R^2 among the models was 0.5 for the model using hours slept, and the highest R^2 was 0.79 with the model using conflicts on social media. This again shows that the number of conflicts has the strongest correlation with the mental health score.



Multiple Linear Regression

Results

Since there is a clear linear trend occurring (parallel lines), the model fails to depict a non-linear relationship, so a non-linear regression model would better capture our data. Ideally, residuals should be scattered around 0.

The model is unable to learn generalizable patterns from our data. There may also be unidentified variables affecting the data and causing these limitations.

Values

Regression Results:

R^2: 0.823

Explanation: Close to 1, most of the variability is explained by the model.

F-statistic: 1087

Probability: 4.33×10^{-263}

Explanation: Model explains a significant amount of our variance in relation to unexplained error.

Highest coefficient value is 0.7983 for conflicts over social media, indicates conflicts the strongest relationship to mental health.

Predictions from Correlation Matrix (covariance):

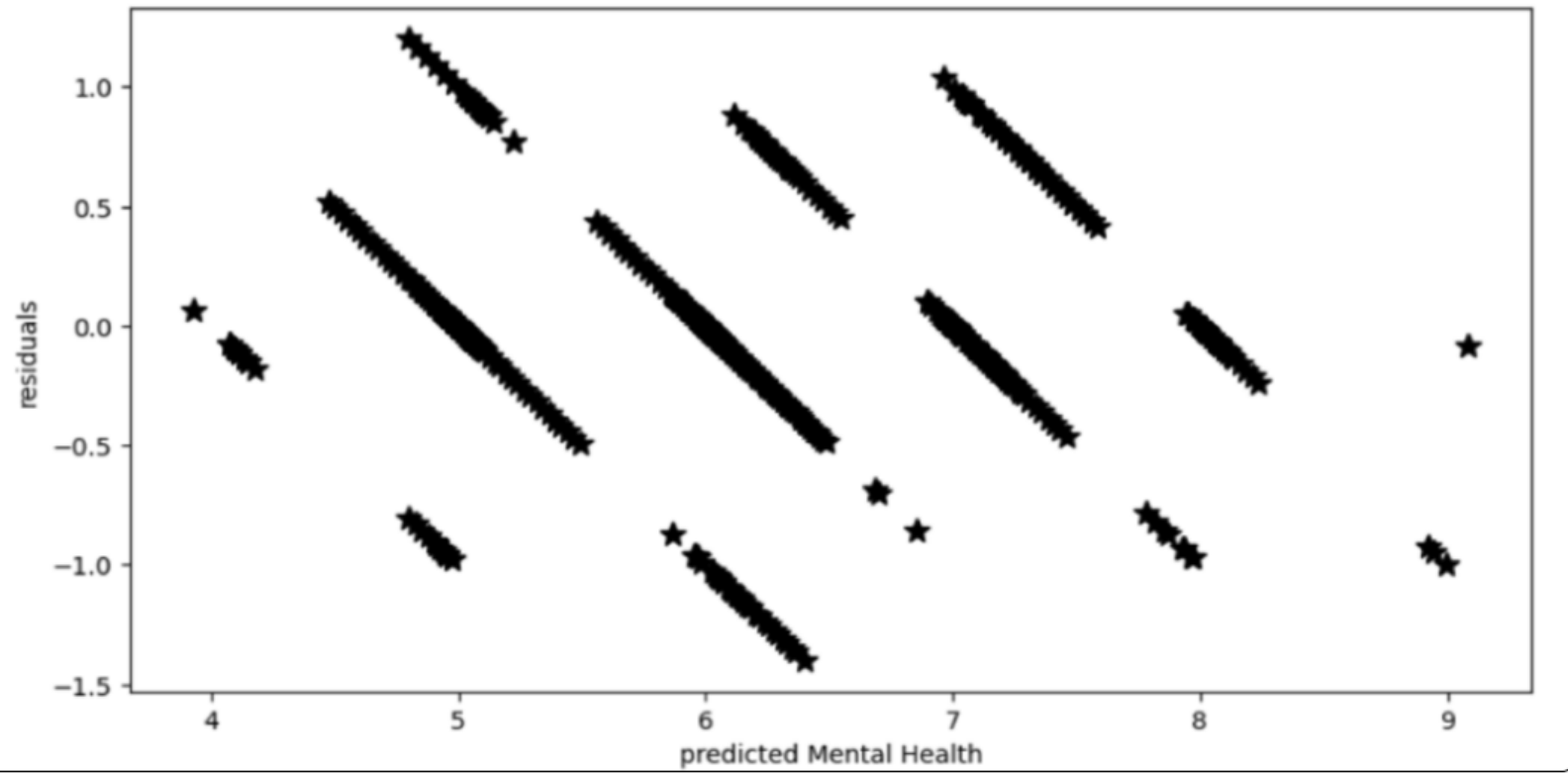
Conflicts: 0.893572

Conflicts and daily usage: 0.804582

Daily usage and sleep hours: 0.790582

Conflicts and sleep hours: 0.677266

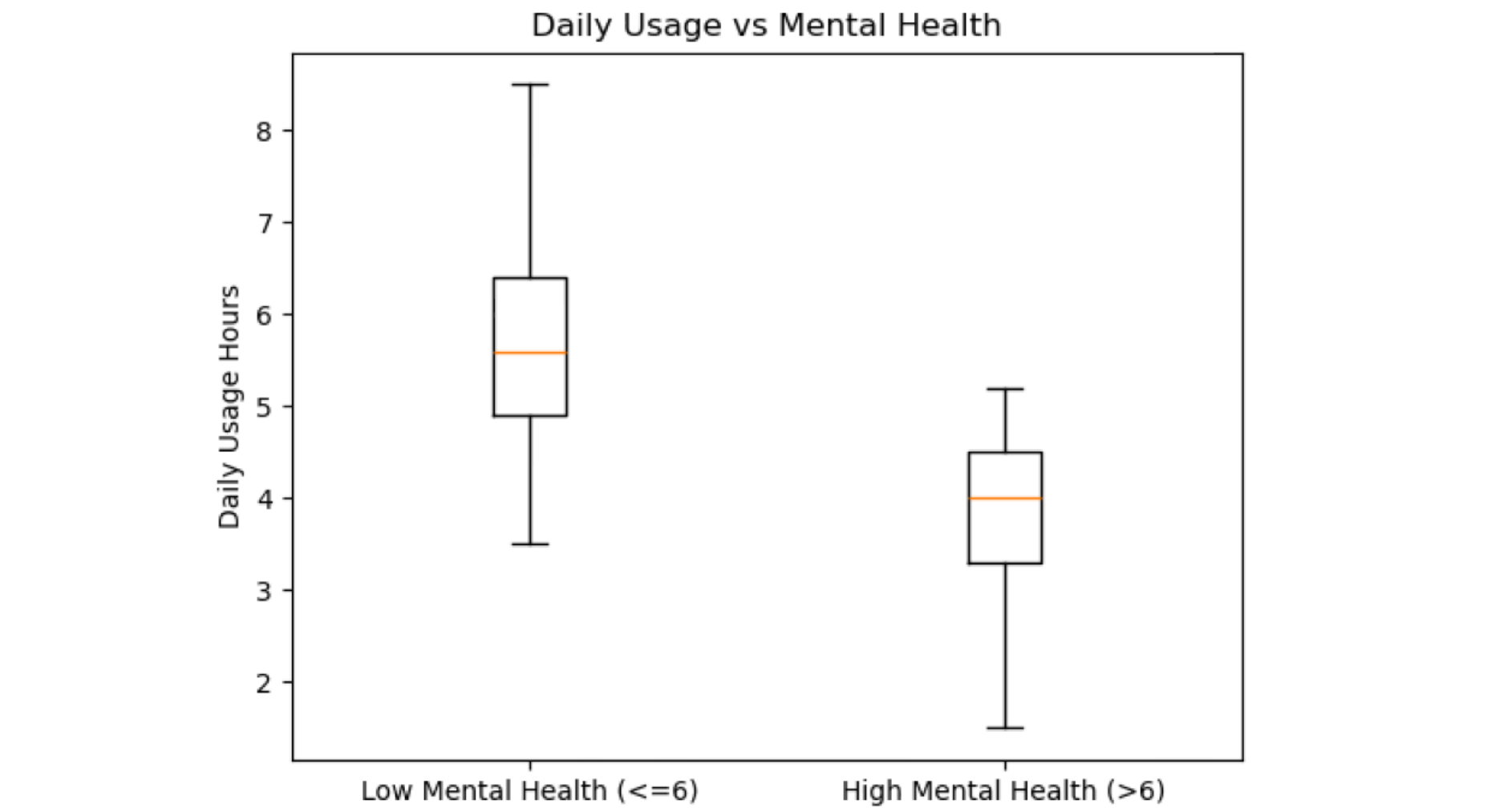
From these covariance values, we observe conflicts alone having the most significance on mental health, and conflicts and daily usage having the most significant combined impact. In comparison to the simple linear regression model, conflicts vs mental health was consistently the most statistically significant and held the strongest correlation in both models. Although, neither simple nor multiple regression were good fits for our dataset, and a non-linear regression model would fit it much better.



Logistic Regression

Goal

Predict mental health based on student habits with 1 being a high mental health score and 0 being low



The box plot shows the general trend of more usage hours leading to lower mental health

Results

Conflicts: Strongest predictor - Beta = -6.32, z = -7.57, p < 0.001
Sleep: Significant positive predictor - Beta = 2.41, z = 3.74, p < 0.001
Daily Usage: Not statistically significant - p = 0.865
R-squared Value = 0.6787 (a good amount of the variability is explained by the model)

Interpretation

Our results show that the only 2 predictors that are contributing are conflicts and sleep. The logistic regression model seems to suggest that the quality of social media interactions and amount of personal rest are better predictors than the quantity of time spent on apps. This makes sense because these are more nuanced predictors and take into account things like lifestyle. Compared to linear regression, logistic regression might lose some detail because our data isn't actually binary and linear regression was able to explain more of the variability.

Conclusion

The Winning Model

For our data, we determined linear regression was better for our data because our response variable was continuous instead of binary.

Future Plans

For future research, we would want to compare more predictor variables and use more results from further surveys from more countries and students. The next steps would be to try other types of modeling that might better model our data like polynomial modeling, ridge regression, and decision tree regression.

What did we learn?

We learned a lot about linear regression, logistic regression, Python coding, and making predictions based on regressions. Personally, I enjoyed the coding and I really enjoyed the group aspect. It was great to connect with people and work together for a common goal.

Improvements to Module 1

I would like to spend a little more time on logistic regression. I felt like we spent a lot of time on linear regression and not as much on logistic.